

Discovery of a Relativistic Sub-Millimeter Jet in the High-Redshift Quasar J1044+3215

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(Received March 15, 2026; Revised April 22, 2026; Accepted May 10, 2026)

Submitted to ApJL

ABSTRACT

We present sub-millimeter continuum observations of the high-redshift ($z = 6.42$) radio-loud quasar J1044+3215 obtained with the Apex Submillimeter Array. We report the clear detection of a spatially resolved, highly collimated relativistic jet extending over a projected scale of 14.2 kpc. The core-jet morphology displays a steep spectral index ($\alpha = -0.84 \pm 0.05$) between 230 GHz and 345 GHz, characteristic of optically thin synchrotron radiation from accelerated ultra-relativistic electrons. Modeling of the broadband spectral energy distribution reveals a minimum equipartition magnetic field strength of $B_{\text{eq}} \approx 1.8$ mG and a total kinetic jet power of $L_{\text{jet}} \approx 3.5 \times 10^{45}$ erg s⁻¹. These findings demonstrate that powerful, large-scale AGN jets can be established within the first billion years of cosmic time, providing crucial feedback constraints for early supermassive black hole growth and host galaxy co-evolution models.

Keywords: Active galactic nuclei (16) — Quasars (1319) — Relativistic jets (1390) — High-redshift galaxies (734) — Radio continuum emission (1340)

1. INTRODUCTION

Relativistic jets powered by supermassive black holes (SMBHs) play a fundamental role in shaping galactic ecosystems across cosmic epoch (Vance et al. 2023). At high redshifts ($z > 6$), the presence of luminous active galactic nuclei (AGN) with SMBH masses exceeding $10^9 M_{\odot}$ poses severe challenges to accretion physics and black hole seed formation models (Rostova et al. 2024). While sub-millimeter and millimeter surveys have uncovered substantial dust reservoirs and cold gas kinematics in these early hosts, spatially resolved sub-millimeter jet components remain exceedingly rare.

The quasar J1044+3215 was previously identified as a radio-loud AGN at $z = 6.42$ with a 1.4 GHz luminosity density of $L_{1.4\text{GHz}} = 4.2 \times 10^{26}$ W Hz⁻¹. Prior radio observations lacked the angular resolution required to disentangle core emission from extended jet knots.

In this Letter, we present high-angular-resolution sub-millimeter continuum imaging from the Apex Submillimeter Array, delivering the first direct structural map of an extended jet at $z > 6$. Throughout this paper, we adopt a flat Λ CDM cosmology with $\Omega_m = 0.308$, $\Omega_{\Lambda} = 0.692$, and $H_0 = 67.8$ km s⁻¹ Mpc⁻¹, yielding a spatial scale of 5.62 kpc arcsec⁻¹ at $z = 6.42$.

2. OBSERVATIONS AND DATA REDUCTION

Observations of J1044+3215 were conducted using the Apex Submillimeter Array in synthesized extended configuration during October 2025 and January 2026. Data were acquired in Band 6 (230.5 GHz) and Band 7 (345.0 GHz) with total on-source integration times of 4.2 hours and 6.8 hours, respectively.

The array configuration provided baseline lengths extending up to 2.4 km, yielding synthesized beam full-width at half-maximum (FWHM) dimensions of $0.18'' \times 0.14''$ at Band 6 and $0.12'' \times 0.09''$ at Band 7. Phase calibration was maintained through periodic observations of the quasar J1058+2811, while primary flux calibra-